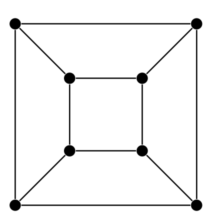
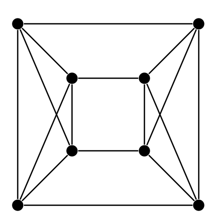
CS105 (DIC on Discrete Structures)
Exercise Problem Set 11

- Attempt *all* questions.
- Apart from things proved in lecture, you cannot assume anything as "obvious". Either quote previously proved results or provide clear justification for each statement.

Part 1

1. Prove or disprove:
   (a) If a maximal trail in a graph is not closed, then its endpoints have odd degree.
   (b) Every Eulerian bipartite graph has even number of edges.
2. From the definition of isomorphism, prove that
$$G \cong H \quad \text{if and only if} \quad \overline{G} \cong \overline{H},$$
where the *complement* of a graph $G = (V, E)$, denoted by $\overline{G}$, is defined as the graph on the same vertex set V such that
$$uv \in E(\overline{G}) \quad \text{if and only if} \quad uv \notin E(G),$$
for all distinct vertices $u, v \in V$.
3. Prove that the graph on the right is isomorphic to the complement of the graph on the left.





4. (a) Construct a simple graph with six vertices that has only one automorphism.
   (b) Construct a simple graph that has exactly three automorphisms.

5. There are 100 circles forming a connected figure on the plane. Can this figure be drawn without lifting the pencil off the paper or drawing any part of any circle twice? Why or why not?

6. Verify that the set of automorphisms of a graph G has the following properties:
   (a) The composition of two automorphisms is an automorphism.
   (b) The identity permutation is an automorphism.
   (c) The inverse of an automorphism is also an automorphism.
   (d) Composition of automorphisms satisfies the associative property.

7. Prove that "every u, v-walk contains a u, v-path".

8. Prove that every n-vertex graph with at least n edges contains a cycle.

Part 2

9. Prove or disprove: If a simple graph G with n vertices has more than $(n-1)(n-2)/2$ edges, then it has only one connected component.

10. Let G be a graph with vertices V and edges E with no self-loops. Show that G has a bipartite sub-graph with at least $E/2$ edges. Give two proofs: one by maximality and contradiction and another by showing an algorithmic construction.

11. (a) Prove that if G is Eulerian and $G' = G - uv$ (where G contains the edge uv), then G' has an odd number of u, v-trails that visit v only at the end. Prove also that the number of the trails in this list that are not paths is even.
    (b) Let v be a vertex of odd degree in a graph. For each edge e incident to v, let $c(e)$ be the number of cycles containing e. Use $\sum_e c(e)$ to prove that $c(e)$ is even for some e incident to v.
    (c) Use part (a) and part (b) to conclude that a nontrivial connected graph is Eulerian if and only if every edge belongs to an odd number of cycles.

12. A mountain range is a polygonal curve from $(a, 0)$ to $(b, 0)$ in the upper half-plane. Hikers A and B begin at $(a, 0)$ and $(b, 0)$, respectively. Prove that A and B can meet by traveling on the mountain range in such a way that at all times their heights above the horizontal axis are the same. (Hint: Define a graph to model the movements)